LECTURE 4

LAST TIME

LECTURE 3 SUMMARY

Forwards are useful to hedge risk. But what is the forward price?

LoOP $\implies$ forward price (no cost of storage): $F_0(1) = p_0(1 + r)$

Expectations and forwards

Forward prices of bonds

Forward exchange rates

THIS TIME

LECTURE 4 OUTLINE

Arb review

Storage costs and convenience yields

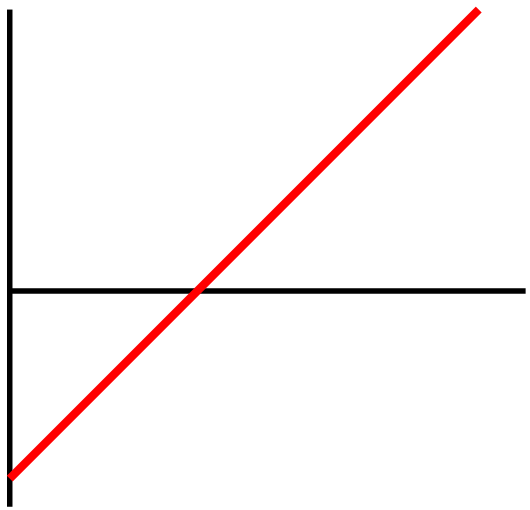
Price (Value) of a forward contract ($\neq$ forward price)

TYPES OF DERIVATIVES

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linear

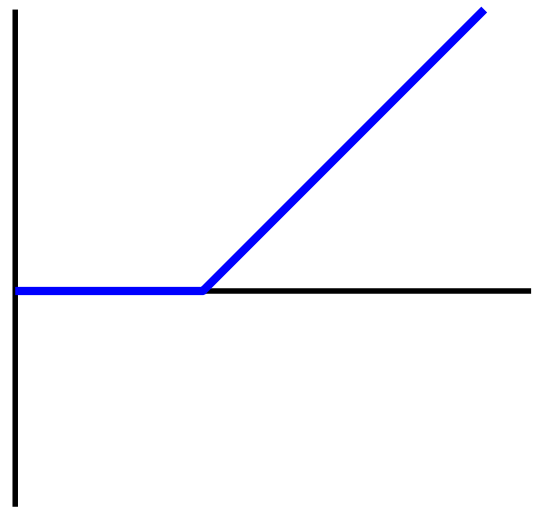
e.g. forwards, futures, swaps



easy (or easier)
priced w/o models
static hedge

non-linear

e.g. options



difficult
require models
dynamic hedge

ARBITRAGE WITH NO UNCERTAINTY

Assume NA; risk free rate $r > 0$

Let V_t denote the value of a portfolio at date $t \in \{0, 1\}$ with $V_0 > 0$

Suppose there is no uncertainty, i.e. V_1 is known today

Can it be that $V_0 > V_1$?

NO ARBITRAGE BOUND

No arbitrage implies "No, arbitrage!"

Proof by contradiction: suppose $V_0 > V_1$

Buy cheap, sell dear: sell portfolio at V_0 and invest at r:

$$\mathrm{CF}_0 = \text{ sale price } - \text{ amount invested } = V_0 - V_0 = 0$$

$$\mathrm{CF}_1 = - \text{ amount owed } + \text{ investment payoff } > -V_1 + (1+r)V_0 > 0$$

since $r > 0$ and $V_0 > V_1$ by hypothesis

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Arbitrage! (which type?) $\implies V_0 \leq V_1$. Can we do better?

APPLY LoOP

V_1 is a certain payoff

Its payoff is the same (in every state) as a portfolio of V_1 bonds

(Recall "bond" is asset with certain payoff one tomorrow)

Replicate! LoOP says V_0 must be value of portfolio $V_0 = \dfrac{V_1}{1+r}$

Must this be less than V_1 in practice?

INTEREST RATES CAN BE NEGATIVE

Euro area yields: https://www.ecb.europa.eu/stats

US yields: https://www.treasury.gov/resource-center/

BACK TO FORWARDS

ASSUMPTIONS BEHIND $F_0(1) = (1 + r)p_0$

To apply LoOP to price the forward, we assumed no arb

But we also assumed replication was feasible and costless

(Always assume this for financials, but not for commodities)

Assumed underlying traded and "shortable"

Assumed zero cost of storage

financial: no dividends

commodity: no storage cost or "convenience yield"

THEORY OF STORAGE

STORAGE COSTS

Commodities are costly to store

Let c be the cost of storage over $[0, 1]$ paid at $t = 1$

Will this increase or decrease the forward price $F \equiv F_0(1)$?

REPLICATION WITH STORAGE COSTS

Payoff of forward contract

$$X_1 = p_1 - F$$

Payoff of portfolio long commodity, short F bonds

$$p_1 - c - F$$

Replicate forward: long commodity and short $F - c$ bonds

$$V_1 = p_1 - c - \big(F - c\big) = p_1 - F$$

FORWARD PRICING WITH COSTLY STORAGE

By the Law of One Price

$$p_0 - \frac{F - c}{1 + r} = 0$$

or

$$F = p_0(1 + r) + c$$

Higher costs of storage $\iff$ higher forward price

Forward avoids storage costs, increasing relative value

Note: Often use $F = p_0(1 + r - y)$, $y := -c/p_0$, convenience yield/unit

What does this say about forward prices over time?

VALUE OF FORWARDS

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What is forward written at 0 maturing at T worth at $t < T$?

Use present value formula $X_t = \mathrm{PV}_t\big[p_T - F_0(T)\big]$

Trick to write in terms of $F_t(T)$: Add and subtract $F_t(T)$:

$$\begin{aligned} X_t &= \mathrm{PV}_t\big[p_T \underbrace{-F_t(T) + F_t(T)}_{=0} - F_0(T)\big] \\ &= \big(F_t(T) - F_0(T)\big)(1+r)^{-(T-t)} \end{aligned}$$

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Note: X_t can be $\lessgtr 0$

ALTERNATIVE TRICK: SHORT FORWARD AT t

Idea: Adding t-T forward to portfolio at t leaves value unchanged

$$X_t = \mathrm{PV}_t\big[p_T - F_0(T)\big] - \underbrace{\mathrm{PV}_t\big[p_T - F_t(T)\big]}_{=0} = \frac{F_t(T) - F_0(T)}{(1+r)^{T-t}}$$

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Both derivations: trick eliminates randomness, price as bonds

VALUE OF FORWARD PRACTICE

Let $r \equiv 4.5\%$. You enter a 2-year forward on oil in 1/93, when $p_{93} = 20$

What is the forward price?

What is your forward worth in 1/94, when $p_{94} = 15$?

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Solution:

$$F_{93}(95) = p_{93}(1+r)^{95-93} = 20 \times 1.045^2 \approx 21.84$$

$$F_{94}(95) = 15 \times 1.045 \approx 15.6$$

Apply formula: $V_{94} = \dfrac{F_t(T) - F_0(T)}{(1+r)^{T-t}} \approx \dfrac{15.6 - 21.84}{1.045} \approx -5.97$

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Note: $V_{94} < p_{93} - p_{94} = -5$—if p stays const, lose money. Why?

REFRESHER: DISCOUNTING

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Compound interest is interest on interest: $(1+r)^2 = 1+2r+r^2 > 1+2r$

Compounding period $= 1/n$th of unit time: r/n is interest/period

$\uparrow n \implies \uparrow$ interest on interest

For interest rate r and compounding period n, value dollar invested for T:

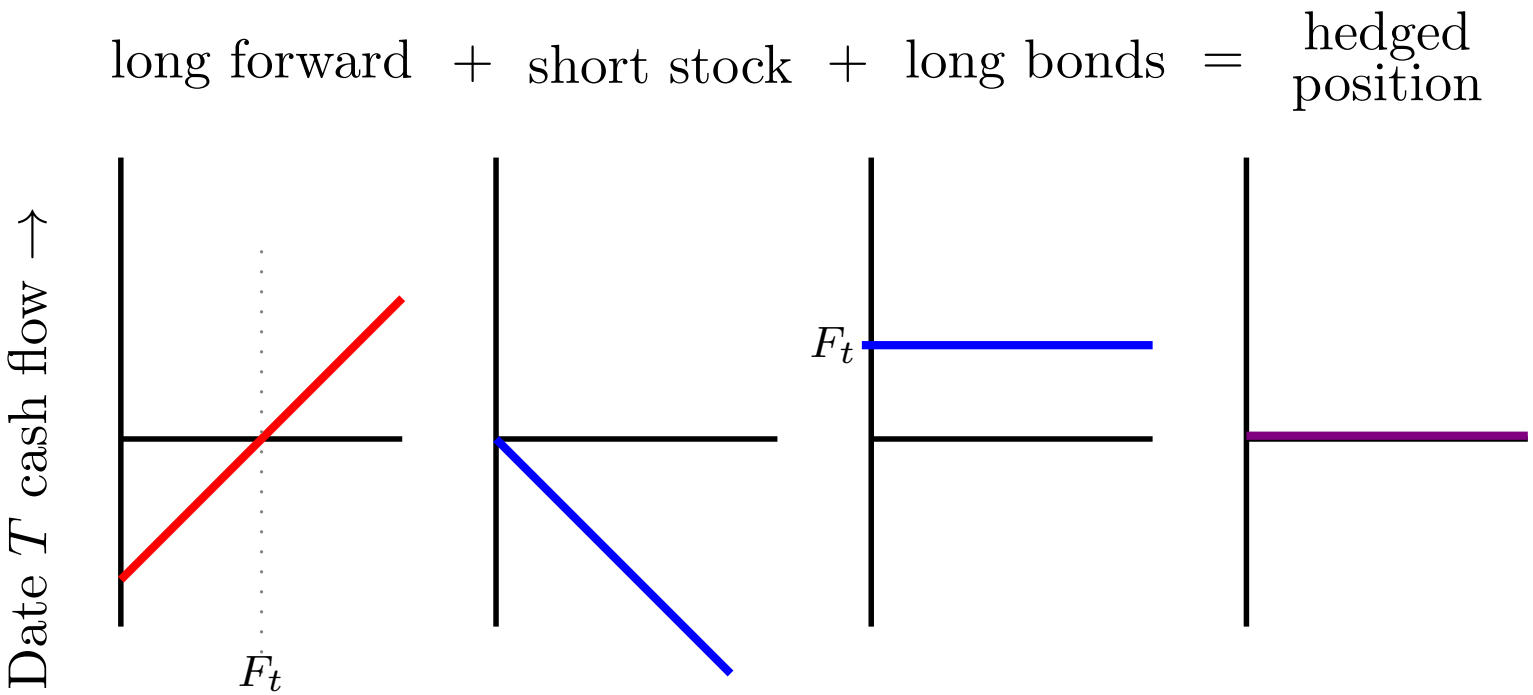
$$\left(1+\frac{r}{n}\right)^{nT} \to e^{rT} \quad \text{as } n \to \infty$$

Typically $n = 365$, but use $n \to \infty$ as approximation for simplicity:

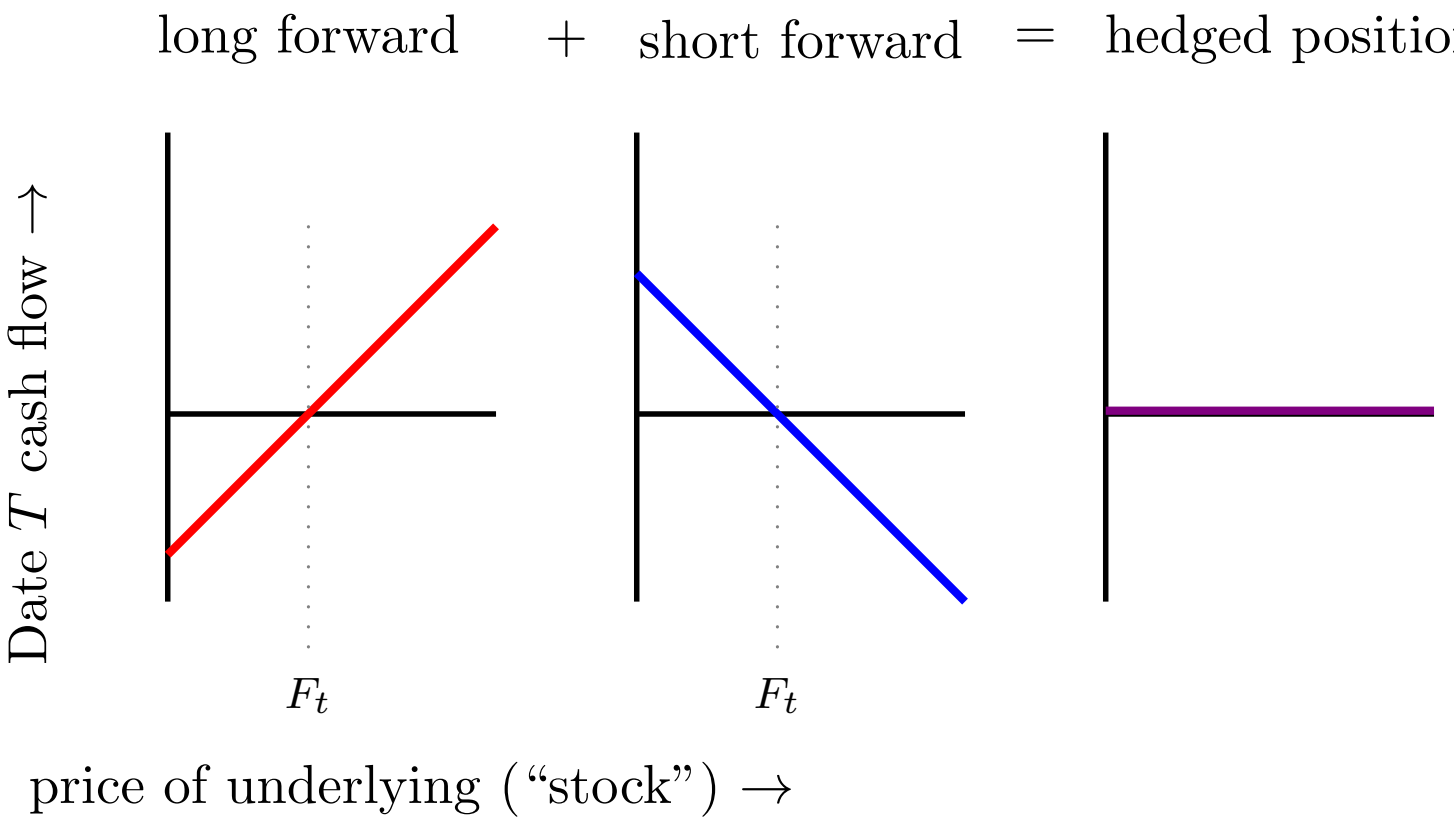
$$\left(1+\frac{5\%}{365}\right)^{365} = 1.051267... \quad \text{vs. } e^{5\%} = 1.0512710...$$

HEDGING FORWARDS

HEDGING FORWARD WITH UNDERLYING



HEDGING FORWARD BACK-TO-BACK



FORWARD CURVE

SLOPE OF THE FORWARD CURVE

With continuous compounding and continuously paid storage costs,

$$F_t(T) = e^{(r-y)(T-t)} p_t$$

(continuous version of $F_t(T) = p_t(1 + r - y)^{T-t}$)

For $r > 0$ and $y < 0$, higher prices for longer dated forwards

If $F_t(T) \uparrow$ in maturity T, curve is normal; if decreasing in T, inverted

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Why could inversion occur?

CONVENIENCE YIELDS

Some securities yield benefits (not costs) to the holder

e.g. stocks pay dividends, bonds coupons

Hence convenience yield y replaced storage cost c

Convenience yields may change and are sometimes a "fudge factor"

RECAP

LECTURE 4: RECAP

Arb review

Storage costs and convenience yields

Value of forward contract

Discounting refresher

Forward curve

FIN